

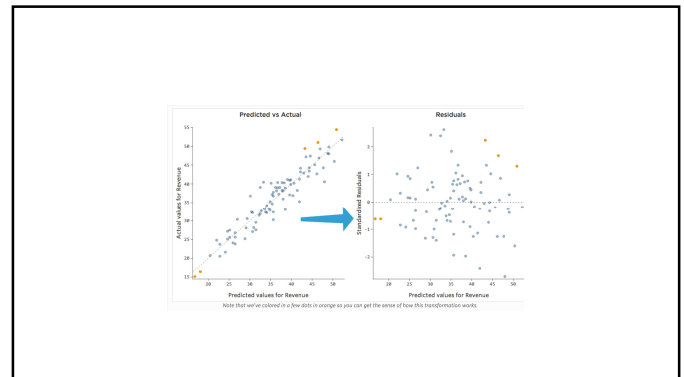


STROJNO UČENJE

7. predavanje

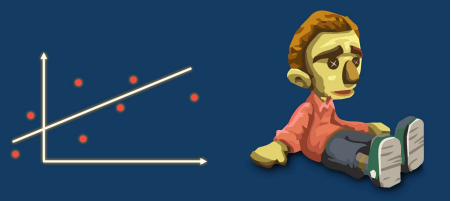
Ocjenjivanje regresora

AI Center Lipik
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7.1 Osnovni (dummy) regresori

DUMMY VARIABLES



Google colab

```
from sklearn import datasets, dummy, metrics # Import datasets, dummy, and metrics
from sklearn import model_selection as skms # Import model_selection as skms
import pandas as pd # Import pandas

diabetes = datasets.load_diabetes()
tts = skms.train_test_split(diabetes.data,
                             diabetes.target, test_size=.25, random_state=42)
(diabetes_train_tts, diabetes_test_tts,
 diabetes_train_tts, diabetes_test_tts) = tts

strategies = ['constant', 'quantile', 'mean', 'median', ]
baseline_args = [{"strategy": s} for s in strategies]
# additional args for constant and quantile
baseline_args[0]['constant'] = 50.0
baseline_args[1]['quantile'] = 0.75
# similar to ch 5, but using a list comprehension
# process a single argument package (a dict)
def do_one(*args):
    baseline = dummy.DummyRegressor(**args)
    baseline.fit(diabetes_train_tts, diabetes_train_tts)
    base_preds = baseline.predict(diabetes_test_tts)
    return metrics.mean_squared_error(base_preds, diabetes_test_tts)
# gather all results via a list comprehension
mses = [do_one(**bla) for bla in baseline_args]
```

7.2 Dodatne mjere za regresiju

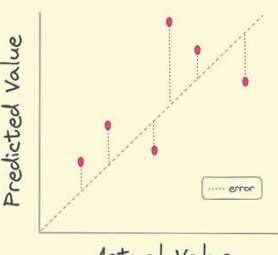
$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i}$$



7.2.1 Stvaranje vlastite metrike procjene

4 Common Regression Metrics

- 1 Mean Squared Error (MSE)
- 2 Root Mean Squared Error (RMSE)
- 3 Mean Absolute Error (MAE)
- 4 R-squared (R²)



```

import numpy as np
from sklearn import metrics, neighbors
from sklearn import model_selection as skms, datasets

def rmse_error(actual, predicted):
    'root-mean-squared-error function'
    # lesser values are better (a < b means a is better)
    mse = metrics.mean_squared_error(actual, predicted)
    return np.sqrt(mse)

def neg_rmse_score(actual, predicted):
    'neg based score function'
    # greater values are better (a < b means b is better)
    return -rmse_error(actual, predicted)

def neg_rmse_score(mod, ftrs, tgt, actual):
    'neg scorer suitable for scoring arg'
    tgt_pred = mod.predict(ftrs)
    return neg_rmse_score(tgt, actual, tgt_pred)

diabetes = datasets.load_diabetes()
km = neighbors.kneighborsregressor(n_neighbors=1)
skms.cross_val_score(km,
                      diabetes.data, diabetes.target,
                      cv=skms.Kfold(5), shuffle=True,
                      scoring=neg_rmse_score)

```

7.2.2 Ostale ugrađene regresijske metrike

1. R^2
2. mean absolute error (MAE)
3. mean squared error (MSE)
4. median absolute error

7.2.3 R^2

- Koeficijent determinacije – koliki postotak varijance naš model objašnjava
- Zadana metrika za regresije

$$SSE_{ours} = \sum_i our_errors_i^2 = \sum_i (our_preds_i - actual_i)^2$$

$$SSE_{mean} = \sum_i mean_errors_i^2 = \sum_i (mean_preds_i - actual_i)^2 = \sum_i (mean - actual_i)^2$$

$$R^2 = 1 - \frac{SSE_{ours}}{SSE_{mean}}$$



```

import numpy as np

our_preds = np.array([1, 2, 1])
mean_preds = np.array([1.5, 2, 2])
actual = np.array([1, 2, 4])
sse_ours = np.sum((our_preds - actual)**2)
sse_mean = np.sum((mean_preds - actual)**2)
r_2 = 1 - (sse_ours / sse_mean)
print("rješenje r2: {}".format(r_2))

```

- Primjer:
- $SSE_{ours}=2500$, $SSE_{mean}=10000$
- $R^2=1-(2500/10000)=1-0.25=0.75$

$$R^2 = 1 - \frac{SSE_{ours}}{SSE_{mean}}$$

$$R^2 = 1 - \frac{\frac{SSE_{ours}}{n}}{\frac{SSE_{mean}}{n}} = 1 - \frac{MSE_{ours}}{MSE_{mean}}$$

$$R^2 = 1 - \frac{SSE_{ours}}{SSE_{mean}} = 1 - \frac{MSE_{ours}}{MSE_{mean}}$$

$$1 - R^2 = \frac{SSE_{ours}}{SSE_{mean}} = \frac{MSE_{ours}}{MSE_{mean}}$$

$$\frac{1 - R_{M1}^2}{1 - R_{M2}^2} = \frac{\frac{MSE_{M1}}{MSE_{mean}}}{\frac{MSE_{M2}}{MSE_{mean}}} = \frac{MSE_{M1}}{MSE_{M2}}$$

Problemi sa sklearn R²



```

import numpy as np

our_preds = np.array([1, 2, 3])
mean_preds = np.array([1.5, 2, 2.5])
actual = np.array([2, 3, 4])
sse_ours = np.sum((our_preds - actual)**2)
sse_mean = np.sum((mean_preds - actual)**2)
r2 = 1 - (sse_ours / sse_mean)
print("Pjelice r2: %.4f" % r2)

# Pjelice r2: 0.40

from sklearn.metrics import r2_score
from sklearn.dummy import DummyRegressor
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split

diabetes = load_diabetes()
diabetes_data = diabetes.data
diabetes_target = diabetes.target

diabetes_train, diabetes_test, diabetes_train_target, diabetes_test_target = train_test_split(
    diabetes_data, diabetes_target, test_size=0.2, random_state=42)

# Pjelice r2: 0.40

```

```

print(base_r2_sklearn)
print(1 - (sse_base_preds / sse_mean_test))

-0.014016723490579253
-0.014016723490579253

```

Problemi sa sklearn R²

$$R^2 = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{baseline}}}$$

$$\text{MSE}_{\text{model}} = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$\text{MSE}_{\text{baseline}} = \frac{1}{n} \sum (y_i - \bar{y})^2$$

$\bar{y}$ je prosjek iz test skupa, a ne iz treniranog modela ili treniranog skupa.

scikit-learn računa R^2 kao normalizirani MSE, gdje se baseline model temelji na prosjeku testnog skupa – što nije reprezentativno za stvarne uvjete treniranja i predikcije.

7.3 Rezidualne plohe

	predicted	actual	error
example			
0	4	3	1
1	2	5	-3
2	9	7	2



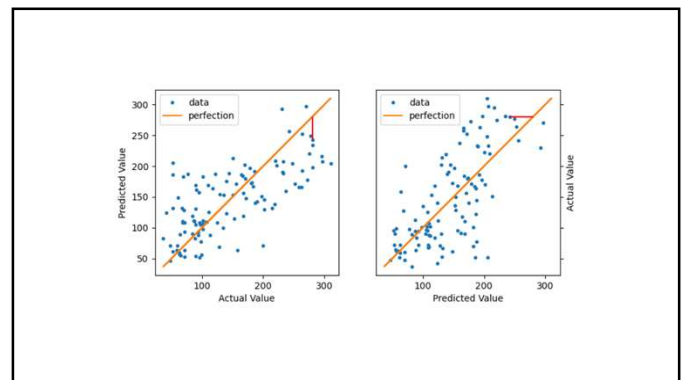
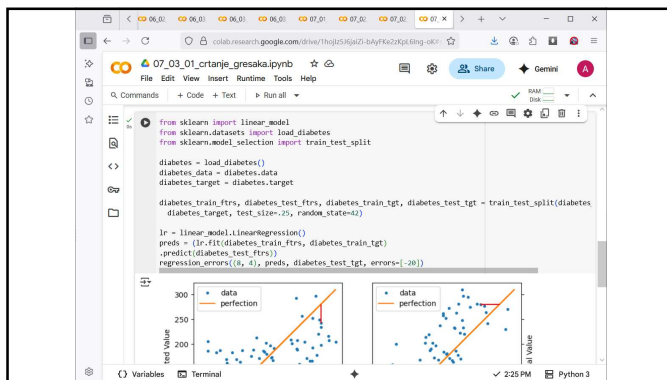
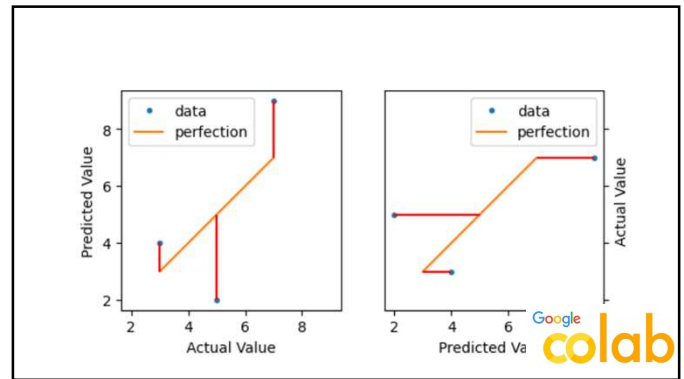
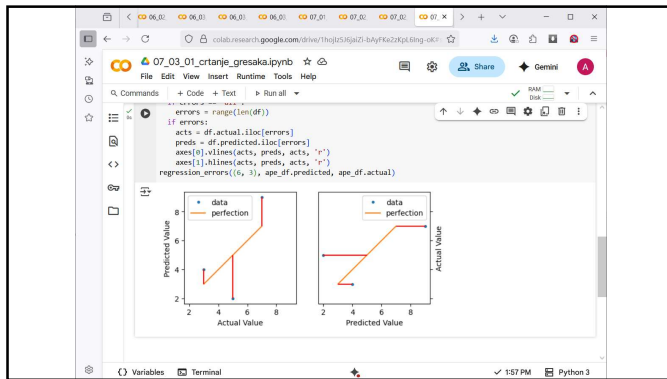
```

import numpy as np
import pandas as pd

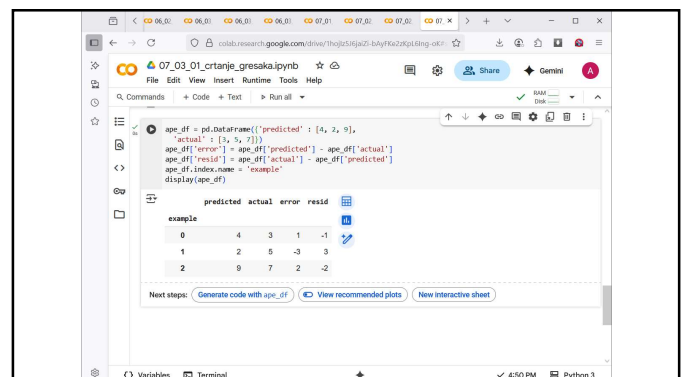
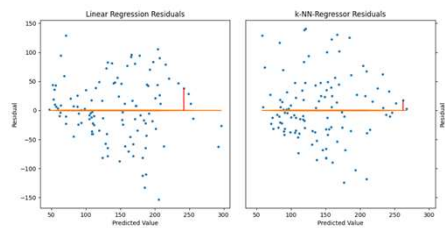
ape_df = pd.DataFrame({'predicted': [4, 2, 9],
                      'actual': [3, 5, 7]})
ape_df['error'] = ape_df['predicted'] - ape_df['actual']
ape_df.index.name = 'example'
display(ape_df)

# predicted actual error
# example
# 0      4      3      1
# 1      2      5     -3
# 2      9      7      2

```

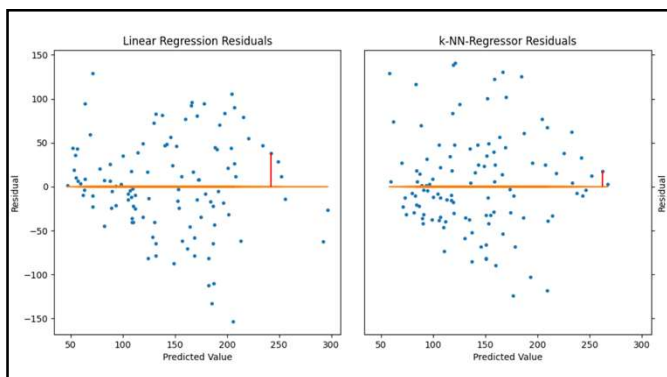
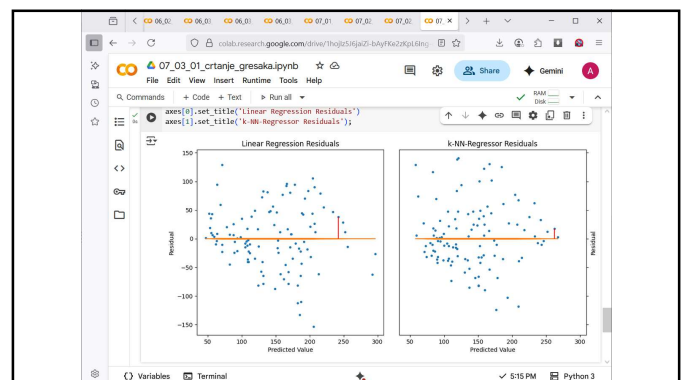
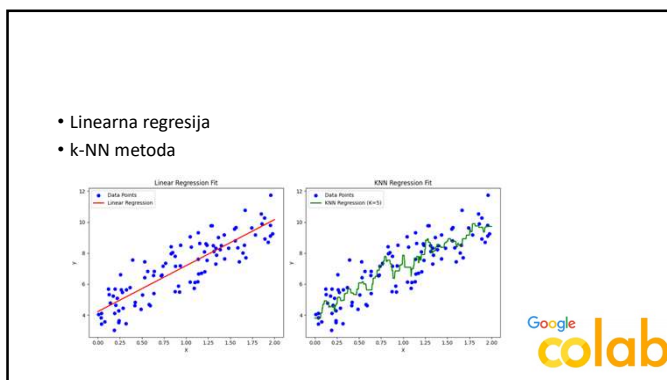
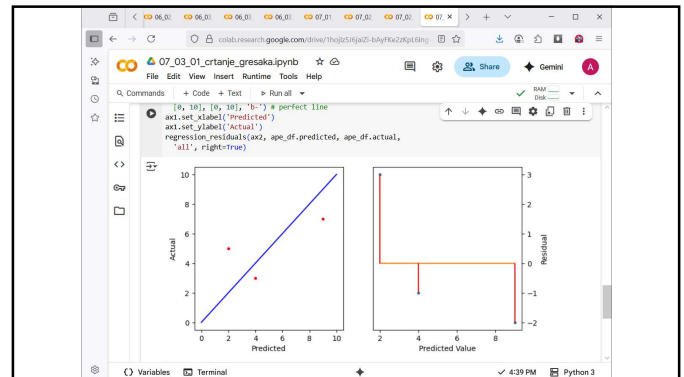


7.3.2 Plohe reziduala



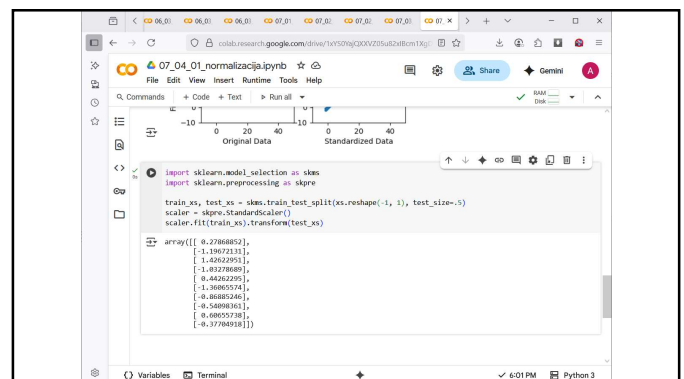
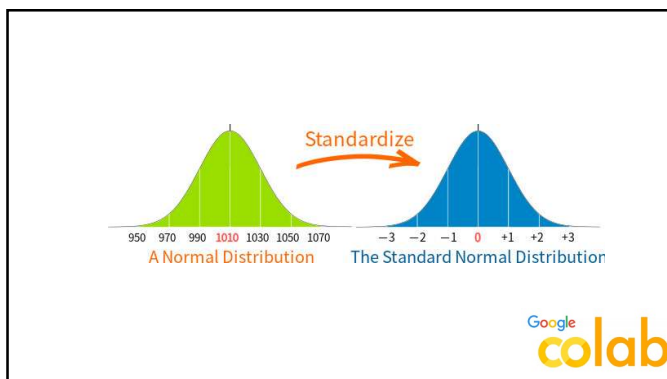
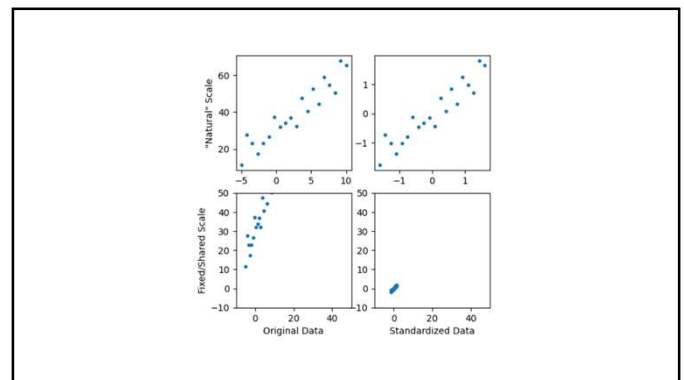
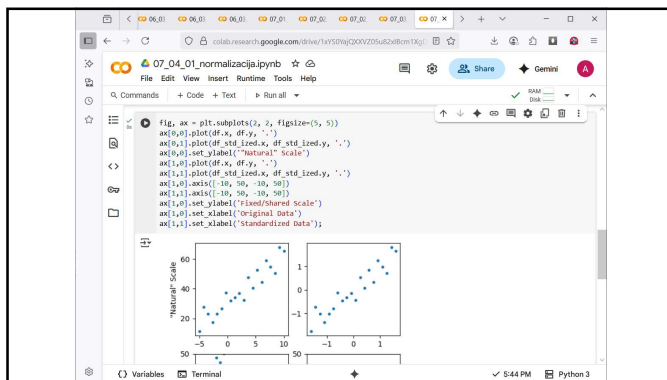
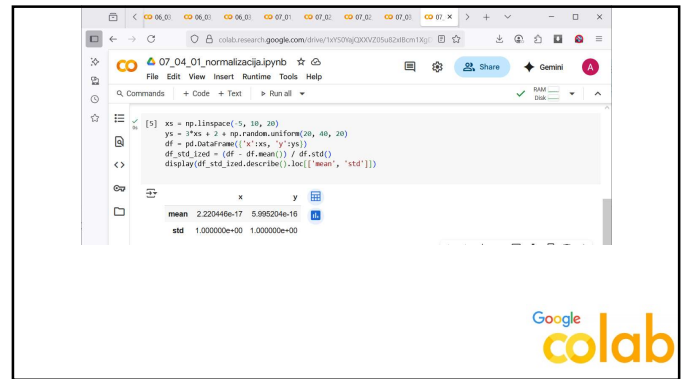
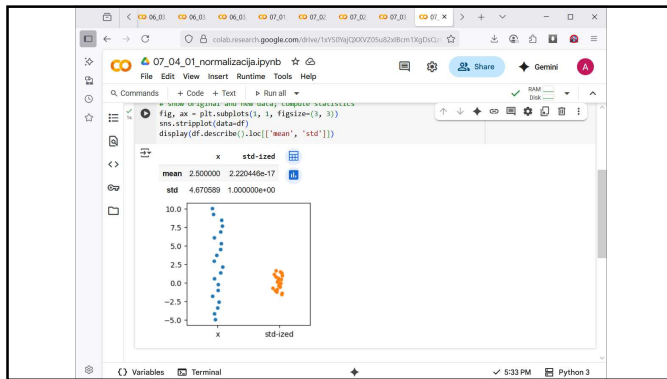
	predicted	actual	error	resid
example				
0	4	3	1	-1
1	2	5	-3	3
2	9	7	2	-2

Google colab

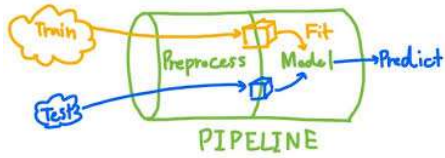


7.4. Normalizacija i standardizacija

- normalizacija je proces uzimanja različitih mjerenja i njihovog stavljanja na izravnu usporedivu osnovu
 - (1) podešavanje središta podataka
 - (2) podešavanje mjerila podataka
- Standardizacija – poseban oblik normalizacije
 - (1) centriramo podatke oko nule oduzimanjem srednje vrijednosti
 - (2) skaliramo podatke tako da imaju standardnu devijaciju od 1 dijeljenjem sa standardnom devijacijom



Pipes/pipelines

Google
colab

```

from sklearn import linear_model, pipeline

(train_xs, test_xs,
 train_ys, test_ys) = sklearn.train_test_split(xs.reshape(-1, 1),
 ys.reshape(-1, 1),
 test_size=.5)
scaler = sklearn.preprocessing.StandardScaler()
lr = linear_model.LinearRegression()
std_lr_pipe = pipeline.Pipeline([('scaler', scaler), ('lr', lr)])
std_lr_pipe.fit(train_xs, train_ys)
std_lr_pipe.predict(test_xs)

array([[29.35477171],
       [34.15404081],
       [41.35204142],
       [50.95147557],
       [10.75623959],
       [24.55508666],
       [17.35608695],
       [40.15209049],
       [60.55080972],
       [45.75257406]])
  
```